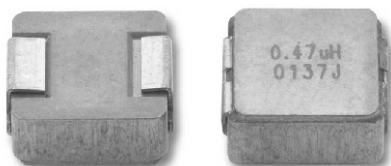


# **IHLP® Commercial Inductors, Low DCR Series**



**DESIGN SUPPORT TOOLS** click logo to get started



| STANDARD ELECTRICAL SPECIFICATIONS                                           |                              |                              |                                                            |                                                        |                      |
|------------------------------------------------------------------------------|------------------------------|------------------------------|------------------------------------------------------------|--------------------------------------------------------|----------------------|
| L <sub>0</sub><br>INDUCTANCE<br>± 20 %<br>AT 100 kHz,<br>0.25 V, 0 A<br>(μH) | DCR<br>TYP.<br>25 °C<br>(mΩ) | DCR<br>MAX.<br>25 °C<br>(mΩ) | HEAT<br>RATING<br>CURRENT<br>DC TYP.<br>(A) <sup>(1)</sup> | SATURATION<br>CURRENT<br>DC TYP.<br>(A) <sup>(2)</sup> | SRF<br>TYP.<br>(MHz) |
| 0.10                                                                         | 2.6                          | 2.9                          | 21.00                                                      | 25.00                                                  | 333                  |
| 0.22                                                                         | 3.5                          | 3.9                          | 21.00                                                      | 14.50                                                  | 170                  |
| 0.33                                                                         | 4.5                          | 5.0                          | 16.50                                                      | 9.00                                                   | 125                  |
| 0.47                                                                         | 5.4                          | 6.0                          | 14.00                                                      | 9.00                                                   | 88                   |
| 1.0                                                                          | 10.0                         | 11.0                         | 10.00                                                      | 6.50                                                   | 50                   |
| 1.5                                                                          | 17.1                         | 18.5                         | 7.50                                                       | 7.00                                                   | 40                   |
| 2.2                                                                          | 22.5                         | 25.0                         | 6.75                                                       | 5.50                                                   | 37                   |
| 3.3                                                                          | 36.4                         | 40.4                         | 5.50                                                       | 7.00                                                   | 29                   |
| 4.7                                                                          | 54.0                         | 60.0                         | 4.50                                                       | 5.20                                                   | 22                   |
| 5.6                                                                          | 63.0                         | 70.6                         | 4.25                                                       | 3.50                                                   | 20                   |
| 10.0                                                                         | 122.1                        | 131.9                        | 2.75                                                       | 2.25                                                   | 17                   |
| 22.0                                                                         | 260.0                        | 270.0                        | 1.90                                                       | 1.70                                                   | 10.7                 |

## **Notes**

- All test data is referenced to 25 °C ambient
- Operating temperature range -55 °C to +125 °C
- The part temperature (ambient + temp. rise) should not exceed 125 °C under worst case operating conditions. Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.
- Rated operating voltage (across inductor) = 40 V
- (1) DC current (A) that will cause an approximate ΔT of 40 °C
- (2) DC current (A) that will cause L<sub>0</sub> to drop approximately 20 %

| DESCRIPTION    |                  |                      |              |                                |  |
|----------------|------------------|----------------------|--------------|--------------------------------|--|
| IHLP-2020CZ-11 | 4.7 μH           | ± 20 %               | ER           | e3                             |  |
| MODEL          | INDUCTANCE VALUE | INDUCTANCE TOLERANCE | PACKAGE CODE | JEDEC® LEAD (Pb)-FREE STANDARD |  |

| GLOBAL PART NUMBER |   |   |   |      |   |   |   |   |   |              |   |                  |   |   |      |        |   |
|--------------------|---|---|---|------|---|---|---|---|---|--------------|---|------------------|---|---|------|--------|---|
| I                  | H | L | P | 2    | 0 | 2 | 0 | C | Z | E            | R | 4                | R | 7 | M    | 1      | 1 |
| PRODUCT FAMILY     |   |   |   | SIZE |   |   |   |   |   | PACKAGE CODE |   | INDUCTANCE VALUE |   |   | TOL. | SERIES |   |

**PATENT(S):** [www.vishay.com/patents](http://www.vishay.com/patents)

This Vishay product is protected by one or more United States and international patents.

## **FEATURES**

- Shielded construction
- Lowest DCR/μH, in this package size
- Handles high transient current spikes without saturation
- Ultra low buzz noise, due to composite construction
- Excellent temperature stability for inductance and saturation
- Excellent DC/DC energy storage up to 1 MHz to 2 MHz. Filter inductor applications up to SRF (see “Standard Electrical Specifications” table)
- IHLP design. PATENT(S): [www.vishay.com/patents](http://www.vishay.com/patents)
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)

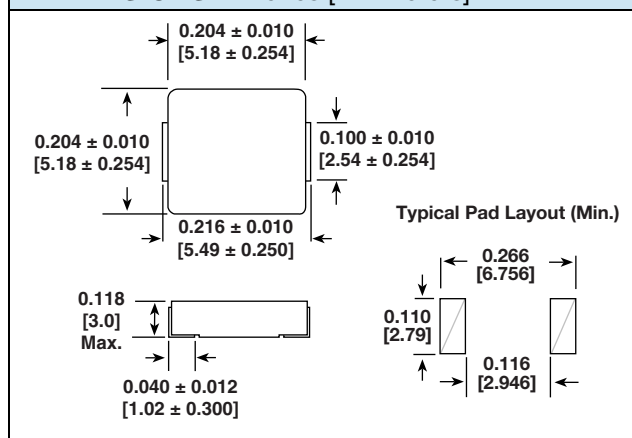


**RoHS**  
COMPLIANT  
HALOGEN  
FREE

## **APPLICATIONS**

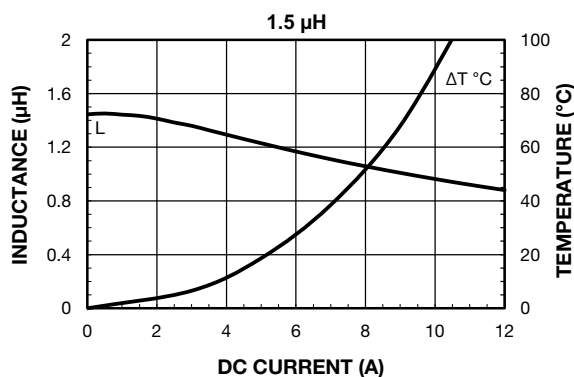
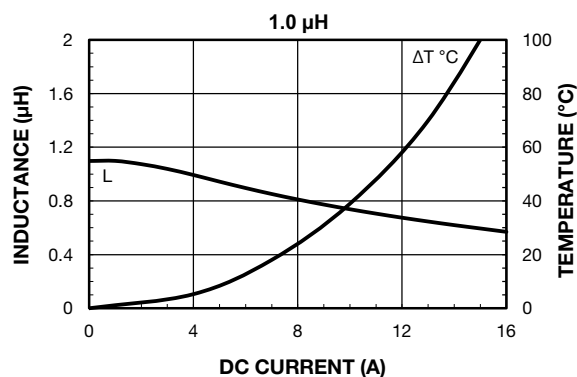
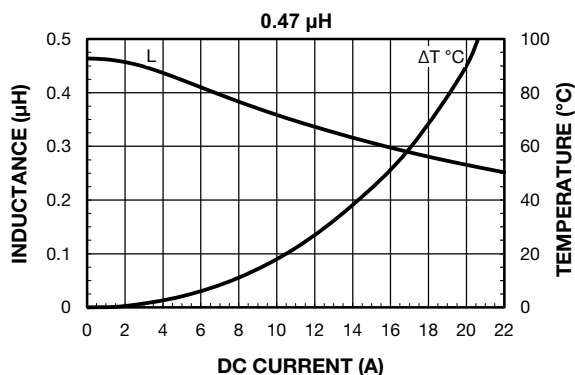
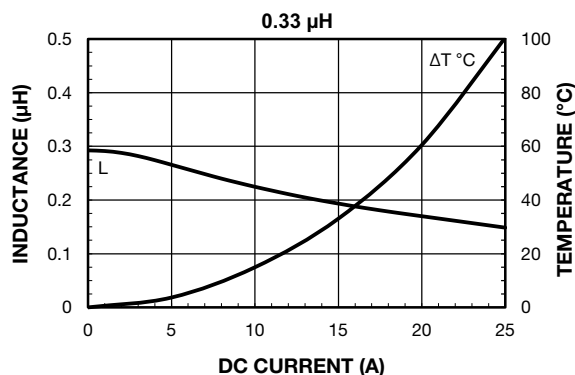
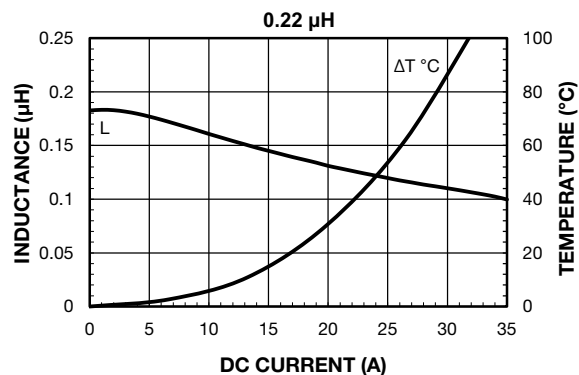
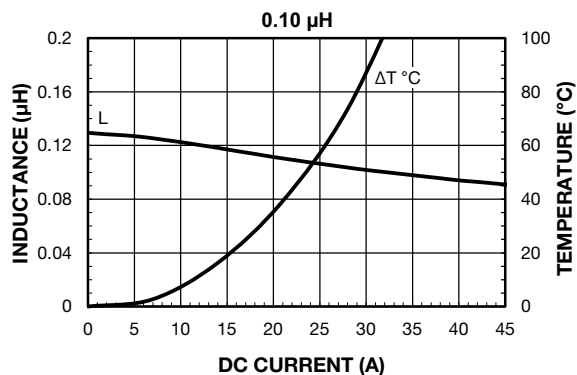
- PDA / notebook / desktop / server applications
- High current POL converters
- Low profile, high current power supplies
- Battery powered devices
- DC/DC converters in distributed power systems
- DC/DC converter for Field Programmable Gate Array (FPGA)

## **DIMENSIONS** in inches [millimeters]



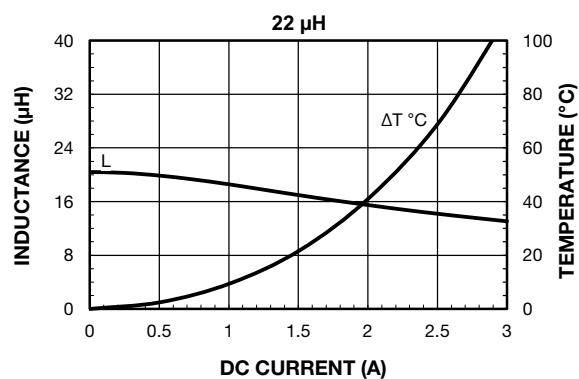
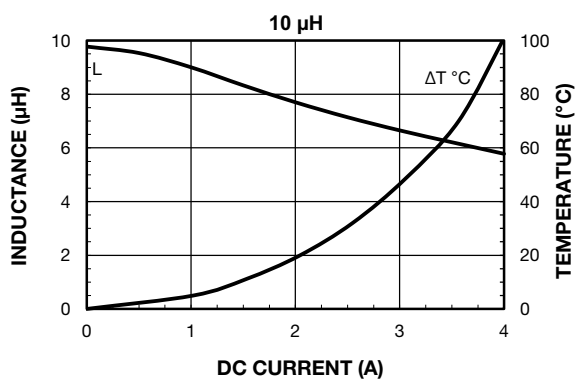
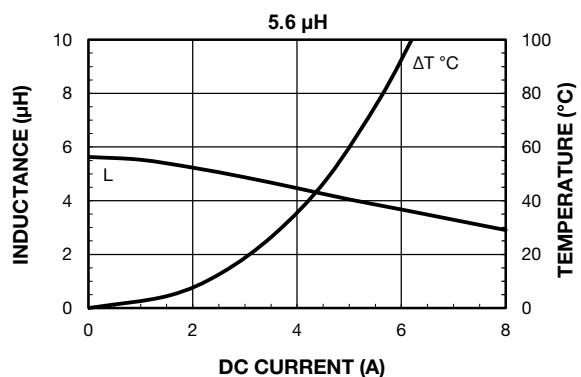
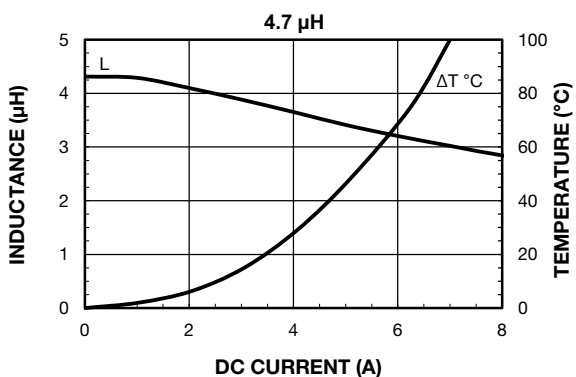
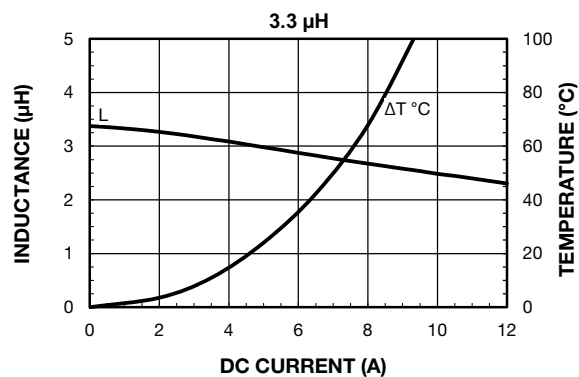
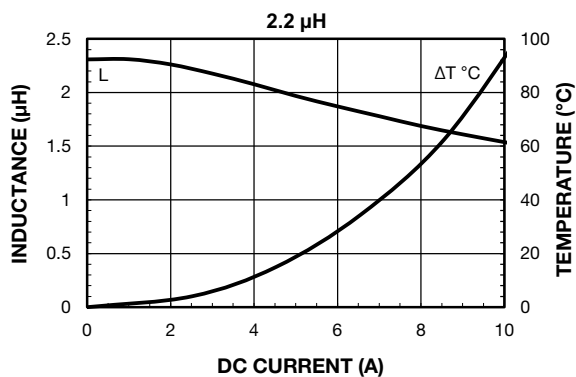


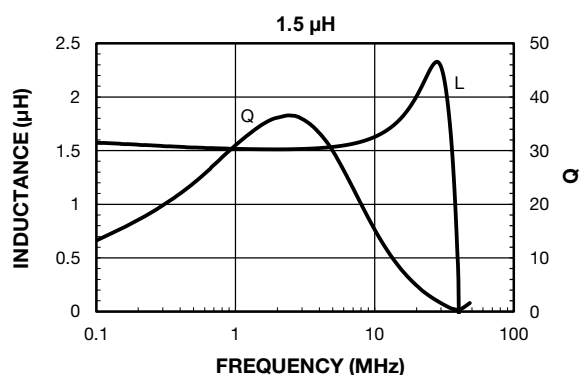
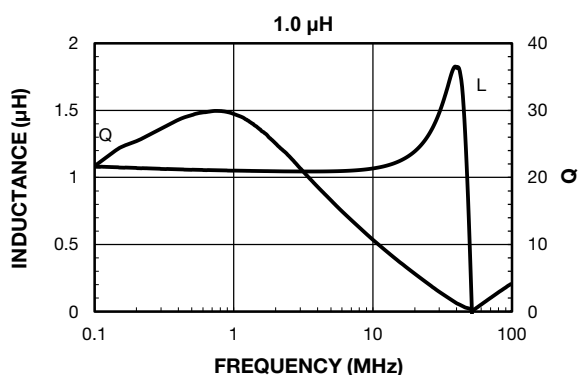
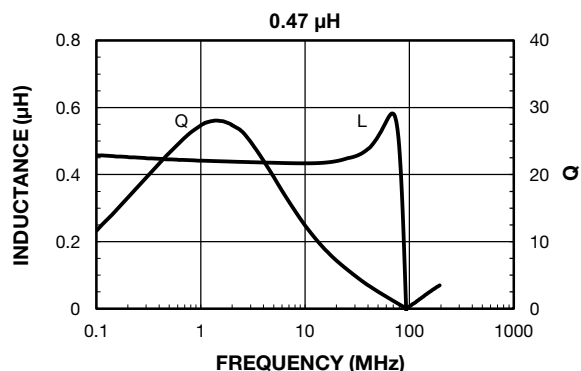
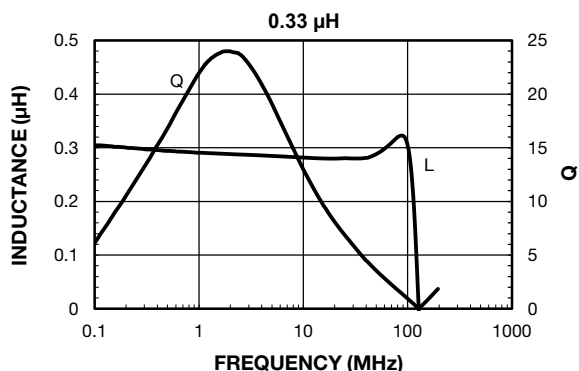
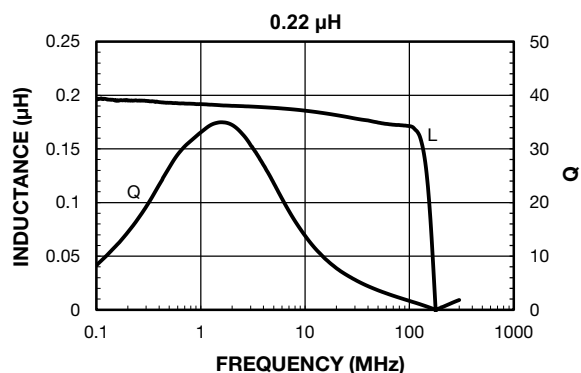
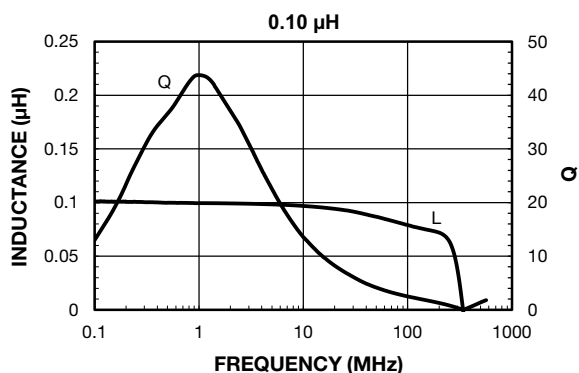
PERFORMANCE GRAPHS

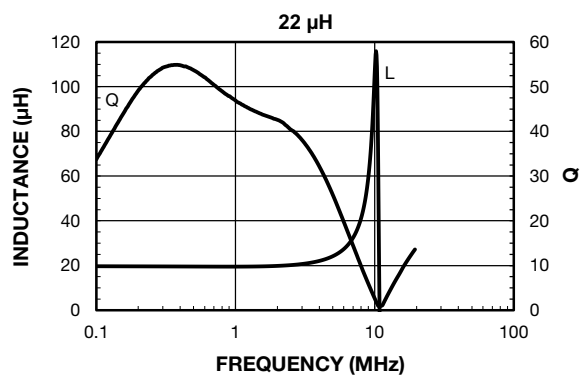
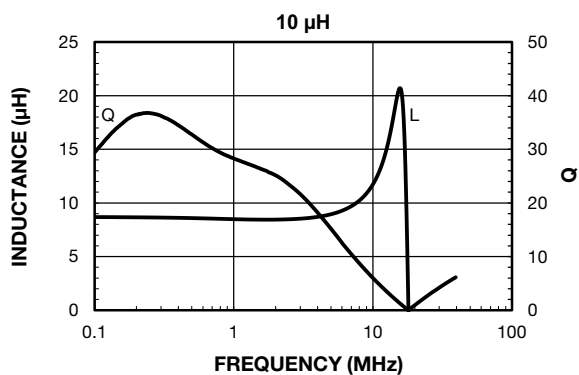
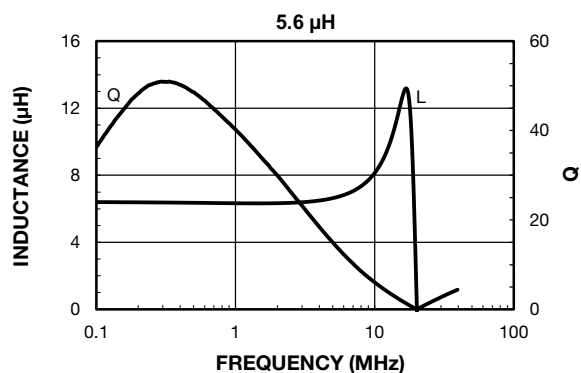
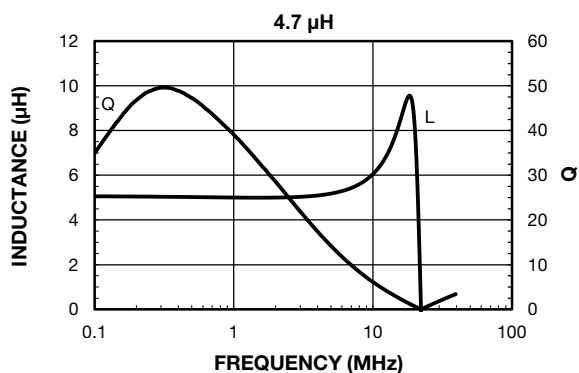
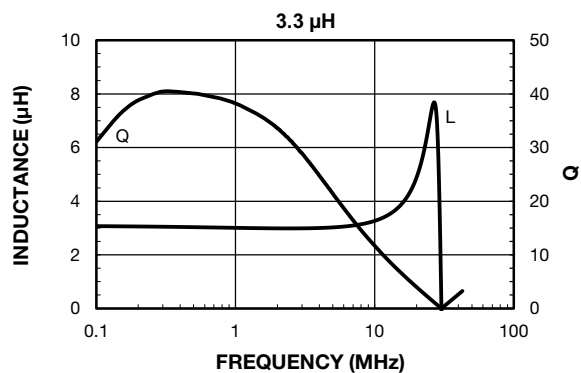
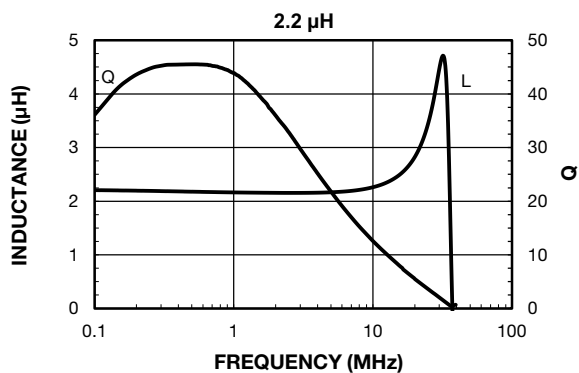




PERFORMANCE GRAPHS



**PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY**


**PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY**




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